

AERIAL OBJECT CLASSIFICATION & DETECTION

Capstone Project Report

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Course:-AI&ML

1. Introduction

This project focuses on aerial image classification and detection using deep learning techniques. The system identifies whether an object in an image is a bird or a drone and also detects its position. This has applications in surveillance, aviation safety, and defense systems.

2. Problem Statement

The objective of this project is to classify aerial images into Bird or Drone and detect the object location using deep learning models.

3. Dataset

The dataset consists of labeled images of birds and drones. It contains two classes with slight imbalance. All images were resized to 224x224 for model training.

4. Data Preprocessing

Data preprocessing included removing invalid images, resizing, normalization, and applying data augmentation such as flipping and zooming to improve model performance.

5. Models Used

Two models were implemented:

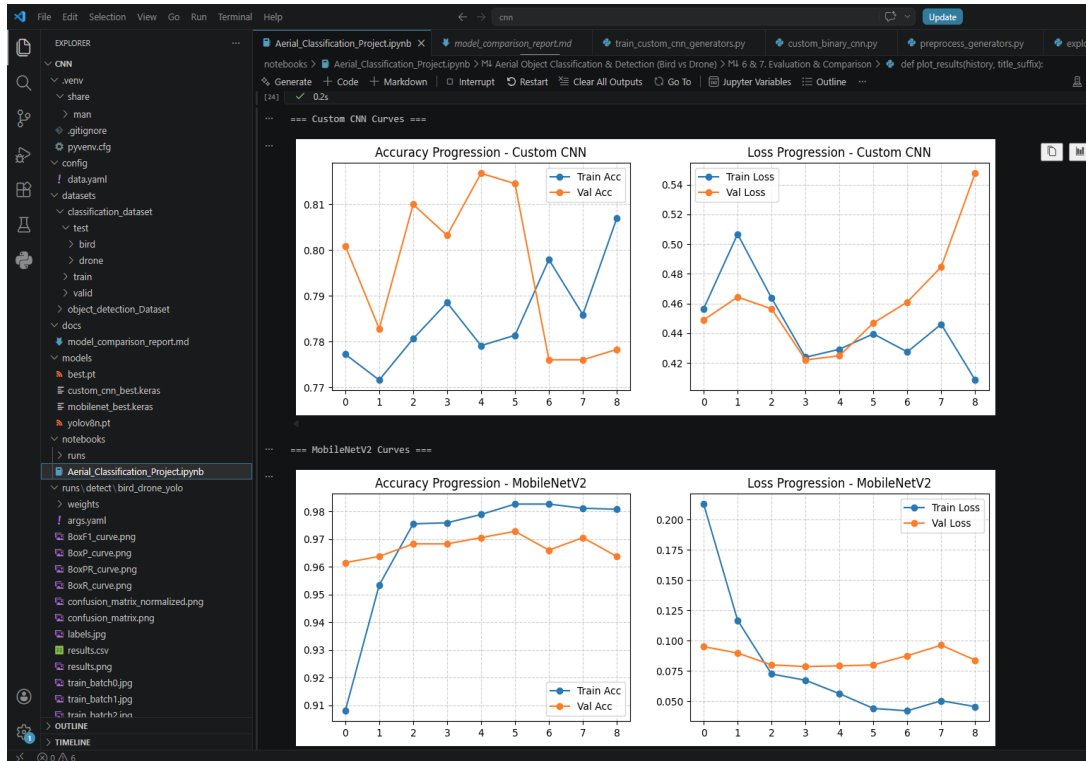
1. Custom CNN: A basic convolutional neural network built from scratch.
2. MobileNetV2: A transfer learning model using pretrained ImageNet weights for better performance.

6. Model Training

Models were trained using augmented data. EarlyStopping was used to prevent overfitting. MobileNetV2 showed faster convergence compared to the custom CNN.

7. Results

The Custom CNN achieved around 76% accuracy, while MobileNetV2 achieved approximately 97% accuracy. MobileNetV2 performed significantly better in terms of generalization and stability.

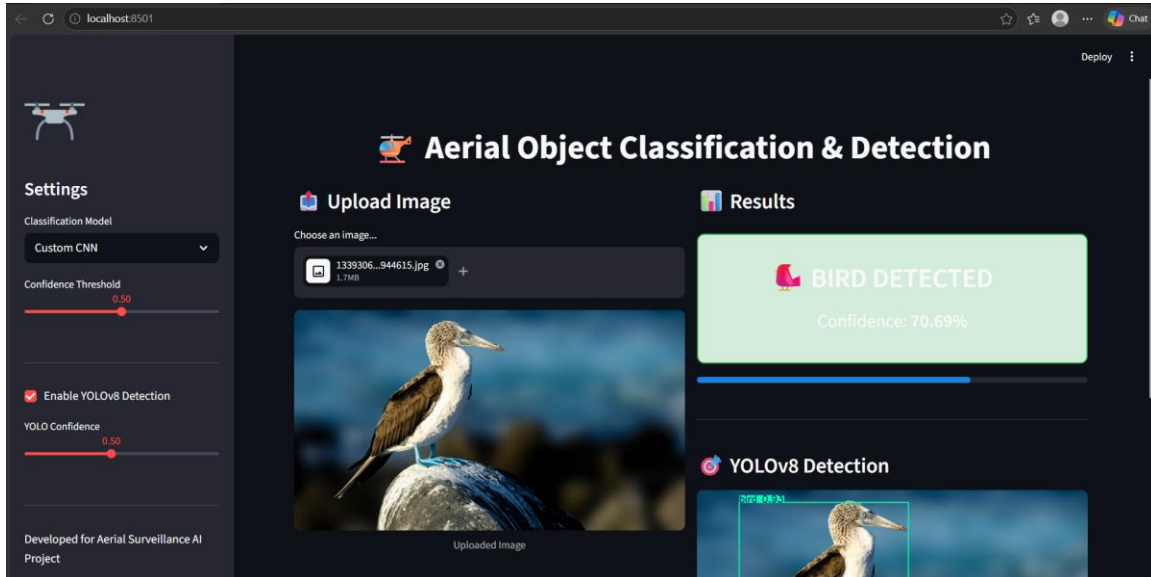


8. Detection using YOLOv8

YOLOv8 was integrated to detect the object location in images. It provides bounding boxes around detected objects with high confidence, improving the system's real-world usability.

9. System Output

The system allows users to upload an image and view results in real-time. It displays the predicted class along with confidence score and shows detection results using bounding boxes.



10. Conclusion

The project demonstrates that transfer learning models like MobileNetV2 outperform custom CNNs for small datasets. The integration of classification and detection makes the system effective for real-world aerial surveillance applications.

11. Future Scope

Future improvements include using larger datasets, improving detection of small objects, and deploying the system for real-time applications.